

Silicon Photonics Mode-Selective 3-dB Power Splitters

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Abstract *Silicon mode-selective 3-dB power splitters based on multimode interference are experimentally demonstrated. They selectively split the TE0 or TE1 mode, allowing the other to pass. Furthermore, AI-based layout correction improves the power imbalance by up to 2.63 dB. ©2026 The Author(s)*

Introduction

Mode-division multiplexing (MDM) in silicon photonics has attracted significant attention as a means of increasing transmission capacity [1,2]. To realize multimode transmission, MDM-compatible passive devices such as mode (de)multiplexers [3] and multimode waveguide bends [4] are required. In recent years, pioneering studies on devices exhibiting mode-dependent operation, such as mode-selective phase shifters [5], have been reported. These devices are expected to serve as key components for applications including mode-selective switching [6] and photonic computing [7].

Mode-selective 3-dB power splitters selectively split only a specific mode, while allowing other modes to pass without splitting. Fig. 1 shows schematics of mode-selective 3-dB power splitters. The TE0-selective splitter splits only the TE0 mode, whereas the TE1-selective splitter splits only the TE1 mode. However, to the best of our knowledge, experimental demonstration of mode-selective 3-dB power splitters in silicon photonics has not yet been reported.

In this paper, we design and experimentally demonstrate these two types of devices based on slotted multimode interference (MMI) couplers. Measurement results show that, over a 60-nm bandwidth, the insertion loss is less than 2.52 dB and the crosstalk is below -10.4 dB. Furthermore, the improvement in power imbalance is confirmed by applying an AI-based layout correction technique that predicts the geometric deviations between designed and fabricated waveguide shapes.

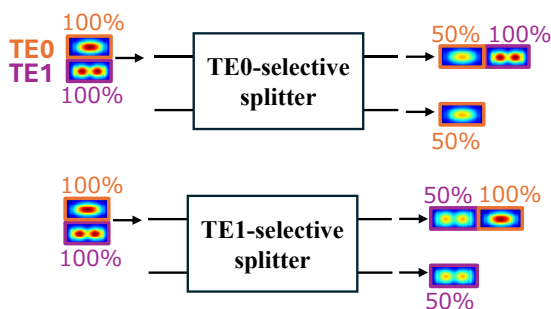


Fig. 1: Schematics of mode-selective 3-dB power splitters.

Structural Design and Simulations

The main challenge in the design is that, in conventional MMI couplers, no imaging position enables TE0- or TE1-selective splitting. To address this issue, we recently proposed that the desired mode-selective splitting can be achieved by introducing two slot regions inside the MMI waveguide [8]. While conventional slotted MMI theory [9,10] is limited to fundamental-mode input, we extend this concept to multimode input operation. In addition, we exploit this design technique to realize mode-selective splitting characteristics that are unattainable with conventional MMI couplers.

Fig. 2 shows schematics of the proposed slotted MMI couplers, where high- and low-refractive-index slots are employed for the TE0- and TE1-selective 3-dB splitters, respectively. In each MMI coupler, with an MMI width of W , the slots and ports are positioned at $\pm W/4$ from the centre of the MMI region. In addition, the two slots have the same refractive index, ensuring symmetry along the y -direction. A key challenge in implementing this device in silicon photonics is the refractive index design of the slot region. The proposed structure requires three regions with different effective refractive indices, namely the core, cladding, and slots. To achieve this, we employ subwavelength grating (SWG) structures. When the period is sufficiently smaller than the wavelength, the SWG behaves as an intermediate-index medium, and

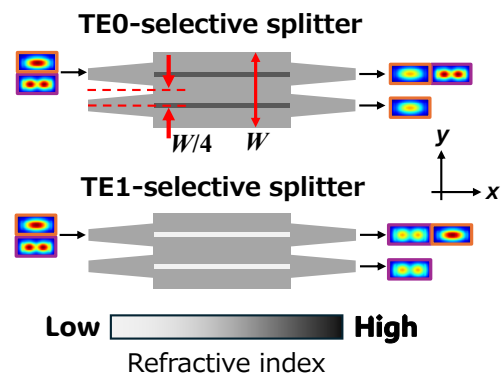


Fig. 2: Schematics of the proposed slotted MMI couplers.

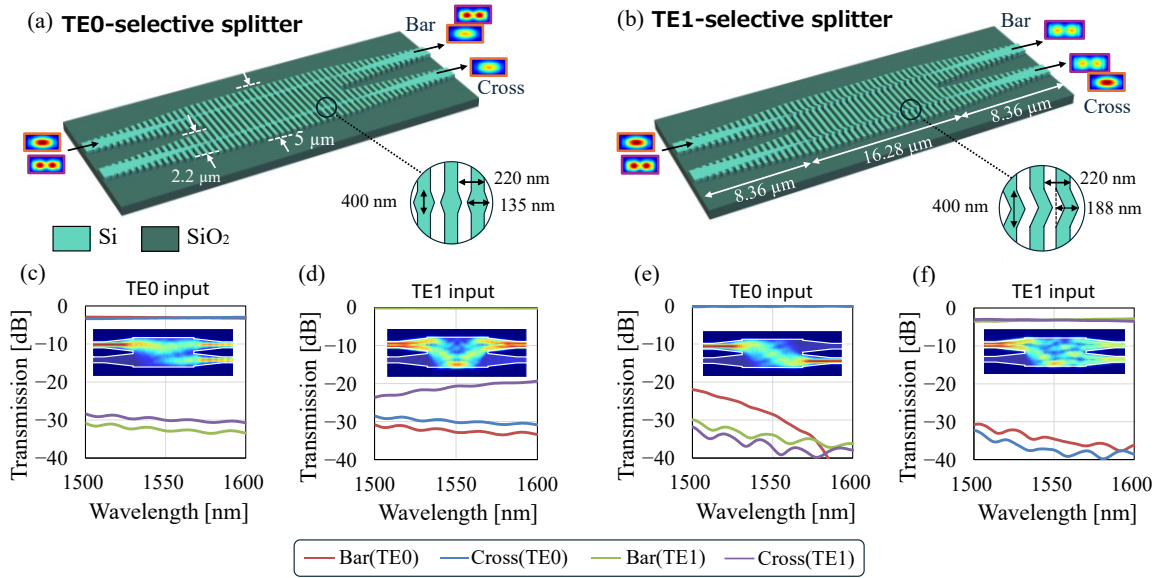


Fig. 3: Structures of (a) TE0-selective splitter, and (b) TE1-selective splitter. Simulated transmission spectra of the TE0-selective splitter for (c) TE0-mode input, and (d) TE1-mode input. Simulated transmission spectra of the TE1-selective splitter for (e) TE0-mode input, and (f) TE1-mode input.

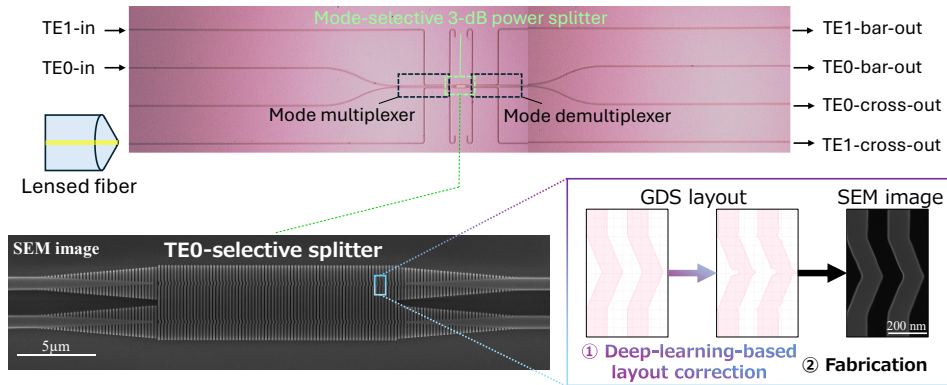


Fig. 4: Overview of the fabricated chip: (top) optical microscope image; (bottom left) SEM image.

thus the effective refractive index can be controlled by adjusting the duty cycle.

Figs. 3(a) and 3(b) show the structures of the respective devices. In both cases, the entire MMI region is composed of SWG structures. In the TE0-selective splitter, high-refractive-index slots are realized by increasing the Si filling factor. In contrast, in the TE1-selective splitter, low-refractive-index slots are achieved—while satisfying the minimum feature size constraint—by exploiting the anisotropy of the SWG structure and bending the grating in the slot region. The structural parameters, except for the slot region, are identical for both devices, which are designed with a compact footprint of $33 \mu\text{m} \times 5 \mu\text{m}$.

Figs. 3(c)–(f) show the transmission spectra obtained by 3D-FDTD simulations. Over the wavelength range of 1500–1600 nm, the maximum insertion loss of the TE0-selective splitter is 0.13 dB for TE0-mode input and 0.18 dB for TE1-mode input, with a maximum crosstalk of –19.4 dB. For the TE1-selective splitter, the maximum insertion loss is 0.21 dB for TE0-mode input and

0.28 dB for TE1-mode input, with a maximum crosstalk of –22 dB. The power imbalance (IB), which characterizes the splitting uniformity, is defined as $IB = 10\log_{10}(T_{\text{Cross}}/T_{\text{Bar}})$ where T_{Bar} and T_{Cross} represent the output powers of the input mode from the bar port and the cross port, respectively. Under this definition, the worst IB is –0.42 dB for the TE0-selective splitter with TE0-mode input, and 0.68 dB for the TE1-selective splitter with TE1-mode input. These results confirm that the proposed devices exhibit selective and uniform splitting characteristics over a wide wavelength range, validating the proposed design principle and its effectiveness.

Experimental results

We fabricated the mode-selective 3-dB power splitters designed in the previous section and performed their experimental evaluation. Fig. 4 shows an overview of the fabricated chip. The devices were fabricated by Applied Nanotools (ANT) foundry. The structures were patterned us-

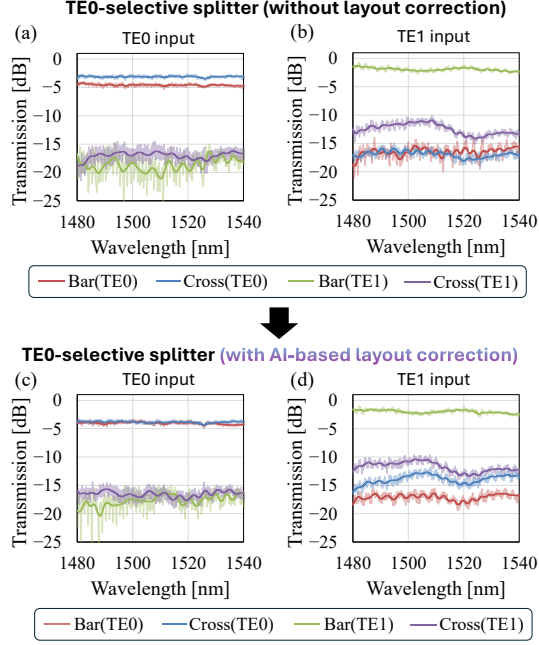


Fig. 5: Transmission spectra of the TE0-selective splitter for (a) TE0-mode input, and (b) TE1-mode input without layout correction, and (c) TE0-mode input, and (d) TE1-mode input with layout correction.

ing electron-beam lithography, followed by anisotropic plasma etching, and finally, a SiO₂ overcladding was deposited. Due to the presence of fine slot structures, ensuring etching uniformity is challenging. Therefore, a deep-learning-based layout correction tool [11] was employed to pre-compensate the pattern variations occurring during fabrication, ensuring the intended shape.

In the measurement setup, the laser light was injected into the chip facet via an inverse-tapered edge coupler with deep trenches and converted into TE0 or TE1 modes using a mode multiplexer. The light transmitted through the proposed MMI coupler was then separated by mode demultiplexers, and the output from each port was measured using an optical power meter. The mode (de)multiplexers were designed based on tapered asymmetric directional couplers [12].

Figs. 5(a)–(d) and Fig. 6(a)–(d) show the transmission spectra of the TE0-selective and TE1-selective splitters, respectively. Panels (a) and (b) correspond to results without layout correction, while panels (c) and (d) correspond to results with layout correction. For the TE0-selective splitter with layout correction, the worst-case *IB* for TE0-mode input was 0.67 dB, the maximum insertion loss was 1.33 dB for TE0 mode and 2.52 dB for TE1 mode, and the maximum crosstalk was –10.4 dB over the measured wavelength range. For the TE1-selective splitter with layout correction, the worst-case *IB* for TE1-mode input was –1.99 dB, the maximum insertion loss was 2.34 dB for TE0 mode and 1.71 dB for TE1 mode,

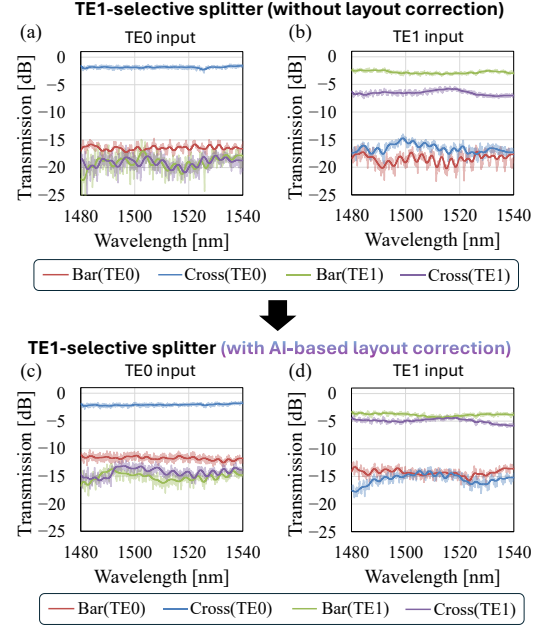


Fig. 6: Transmission spectra of the TE1-selective splitter for (a) TE0-mode input, and (b) TE1-mode input without layout correction, and (c) TE0-mode input, and (d) TE1-mode input with layout correction.

and the maximum crosstalk was –11.2 dB. In both devices, layout correction significantly improved the splitting uniformity. Specifically, for the TE0-selective splitter, *IB* for TE0-mode input improved by an average of 1.28 dB across the measured wavelength range compared to the uncorrected device. Similarly, for the TE1-selective splitter, *IB* for TE1-mode input improved by an average of 2.63 dB.

The central operating wavelength was found to shift to approximately 1510 nm, primarily due to fabrication deviations in the slot structures. As a result, crosstalk and insertion loss were slightly degraded compared to the simulations. Nevertheless, the operating principle of the proposed mode-selective 3-dB power splitters was experimentally validated. The agreement between experiment and simulation could be further improved by optimising the basic SWG parameters (e.g., period and duty cycle).

Conclusions

In this study, we proposed and experimentally demonstrated two types of mode-selective 3-dB power splitters. By employing the slotted MMI theory extended to multimode inputs, we demonstrated that MMI couplers with mode-selective functionality are practically feasible. Furthermore, by applying an AI-based layout correction technique, the splitting uniformity was significantly improved. These results expand the design flexibility for MDM systems.

References

- [1] X. Wu, C. Huang, K. Xu, C. Shu, and H. K. Tsang, "Mode-division multiplexing for silicon photonic network-on-chip," *Journal of Lightwave Technology*, vol. 35, no. 15, pp. 3223–3228, 2017, DOI: [10.1109/JLT.2017.2677085](https://doi.org/10.1109/JLT.2017.2677085)
- [2] L. W. Luo, N. Ophir, C. P. Chen, L. H. Gabrielli, C. B. Poitras, K. Bergmen, and M. Lipson, "WDM-compatible mode-division multiplexing on a silicon chip," *Nature Communications*, vol. 5, no. 1, Art. no. 3069, 2014, DOI: [10.1038/ncomms4069](https://doi.org/10.1038/ncomms4069)
- [3] Y. He, Y. Zhang, Q. Zhu, S. An, R. Cao, X. Guo, C. Qiu, and Y. Su, "Silicon high-order mode (De)multiplexer on single polarization," *Journal of Lightwave Technology*, vol. 36, no. 24, pp. 5746–5753, 2018, DOI: [10.1109/JLT.2018.2878529](https://doi.org/10.1109/JLT.2018.2878529)
- [4] L. H. Gabrielli, D. Liu, S. G. Johnson, and M. Lipson, "On-chip transformation optics for multimode waveguide bends," *Nature Communications*, vol. 3, no. 1, Art. no. 1217, 2012, DOI: [10.1038/ncomms2232](https://doi.org/10.1038/ncomms2232)
- [5] S. M. R. Safaee, K. R. Mojaver, G. Zhang, and O. Liboiron-Ladouceur, "Silicon photonics mode-selective phase shifter," *Journal of Lightwave Technology*, vol. 42, no. 7, pp. 2412–2421, 2024, DOI: [10.1109/JLT.2023.3336589](https://doi.org/10.1109/JLT.2023.3336589)
- [6] S. S. Morrison, S. M. R. Safaee, Z. Ma, R. Rogers, D. Gostimirovic, K. R. Mojaver, and O. Liboiron-Ladouceur, "Enabling ROADM in mode division multiplexing networks with mode-selective switches," *Journal of Optical Communications and Networking*, vol. 17, no. 3, pp. A234–A246, 2025, DOI: [10.1364/JOCN.538478](https://doi.org/10.1364/JOCN.538478)
- [7] L. T. Feng, M. Zhang, X. Xiong, D. Liu, Y. J. Cheng, F. M. Jing, X. Z. Qi, Y. Chen, D. Y. He, G. P. Guo, G. C. Guo, D. X. Dai, and X. F. Ren, "Transverse mode-encoded quantum gate on a silicon photonic chip," *Physical Review Letters*, vol. 128, no. 6, Art. no. 060501, 2022, DOI: [10.1103/PhysRevLett.128.060501](https://doi.org/10.1103/PhysRevLett.128.060501)
- [8] S. Izumisawa, T. Sato, T. Muratsubaki, and K. Saitoh, "Mode-dependent 3-dB power splitters using silicon slotted multimode interference couplers for efficient mode-selective switches," in *2025 30th OptoElectronics and Communications Conference (OECC) and 2025 International Conference on Photonics in Switching and Computing (PSC)*, Sapporo, Japan, 2025, Paper. TuF1-2, DOI: [10.23919/OECC/PSC62146.2025.11109379](https://doi.org/10.23919/OECC/PSC62146.2025.11109379)
- [9] D. M. Mackie and A. W. Lee, "Slotted multimode-interference devices," *Applied Optics*, vol. 43, no. 36, pp. 6609–6619, 2004, DOI: [10.1364/AO.43.006609](https://doi.org/10.1364/AO.43.006609)
- [10] A. Ortega-Moñux, C. Alonso-Ramos, A. Maese-Novo, R. Halir, L. Zavargo-Peche, D. Pérez-Galacho, I. Molina-Fernández, J. G. Wangüemert-Pérez, P. Cheben, J. H. Schmid, J. Lapointe, D. Xu, and S. Janz, "An ultra-compact multimode interference coupler with a sub-wavelength grating slot," *Laser & Photonics Reviews*, vol. 7, no. 2, pp. L12–L15, 2013, DOI: [10.1002/lpor.201200106](https://doi.org/10.1002/lpor.201200106)
- [11] PreFab-Photonics, "PreFab GitHub," <https://github.com/PreFab-Photonics/PreFab>, accessed on Apr. 2, 2026.
- [12] K. Nakamura, T. Fujisawa, T. Sato, and K. Saitoh, "High-tolerance CWDM4 wavelength multiplexer based on 2×2/2×1 MZ filters with polarization multiplexing," *IEICE Electronics Express*, vol. 18, no. 8, pp. 20210110, 2019, DOI: [10.1587/elex.18.20210110](https://doi.org/10.1587/elex.18.20210110)